3. Solve:- $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 8x^2 e^{2x} \sin 2x$

Try Yourself.

[G.S. $y = (C_1 + C_2 x)e^{2x} + e^{2x}(-2x^2 \sin 2x - 4x\cos 2x + 3\sin 2x)$]

linear D.E.

(*) Homogeneous linear D.E. of nth order OR Cauchy-Euler Eqn:

A linear differential equation of the form

$$x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_n y = X \quad \text{——— (1.)}$$

i.e., $$(x^n D^n + a_1 x^{n-1} D^{n-1} + \cdots + a_n) y = X \quad \text{——— (2.)}$$

$a_1, \ldots, a_n$: Constants.

X : either a constant or a function of x, only.

is called a homogeneous linear Differential equation.

or Cauchy-Euler equation.

Working Rule for Solution of eqns of type (1) :-

Method of change of independent variable

$\frac{dy}{dx}$ → dependent variable

dx → independent variable

$\frac{dy}{dx} \longrightarrow \frac{dy}{dz}$

$x \longrightarrow z$

by the relation $x = e^z$

i.e. $\boxed{z = \log x}$

This will transform eqn (1) or (2) to a linear D.E. with Constant Coefficient.

In the last, replace z by $\log x$, in order to get the solution in original variables x and y.

$x = e^z$

$\Rightarrow z = \log x$

$\Rightarrow \boxed{\frac{dz}{dx} = \frac{1}{x}}$

$$\frac{dy}{dx} = \frac{dy}{dz}\frac{dz}{dx} = \frac{1}{x}\frac{dy}{dz}$$

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i.e. $x\frac{dy}{dx} = \frac{dy}{dz}$ or $x\frac{d}{dx} \equiv \frac{d}{dz} \equiv D_1$, where $D_1 \equiv \frac{d}{dz}$

$$\Rightarrow \boxed{xD \equiv D_1}$$

Similarly. $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right)$

$$= -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d^2y}{dz^2}\cdot\left(\frac{dz}{dx}\right)$$

$$= -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d^2y}{dz^2}\left(\frac{1}{x}\right)$$

$$= -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x^2}\frac{d^2y}{dz^2}$$

or $x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dz^2} - \frac{dy}{dz} = D_1^2y - D_1y = (D_1^2 - D_1)y$

$$= D_1(D_1-1)y$$

$$\Rightarrow \boxed{x^2D^2 \equiv D_1(D_1-1)}$$

So, $xD \equiv D_1$

$x^2D^2 \equiv D_1(D_1-1)$

$x^3D^3 \equiv D_1(D_1-1)(D_1-2)$

where $D \equiv \frac{d}{dx}$ and $D_1 \equiv \frac{d}{dz}$.

Exp.

Solve $(x^2D^2 - xD + 2)y = x\log x$ ——(1)

Put $x = e^z$

$\Rightarrow \boxed{z = \log x}$, then eqn (1) becomes

$$(D_1(D_1-1) - D_1 + 2)y = e^z \log e^z$$

i.e. $(D_1^2 - 2D_1 + 2)y = ze^z$

$\Downarrow$ linear DE in y and z with const. coefficient.

A.E. $m^2 - 2m + 2 = 0$

$$\Rightarrow m = \frac{2 \pm \sqrt{4-8}}{2} = \frac{2 \pm 2i}{2} = 1 \pm i$$

So, $C.F. = e^z[C_1\cos z + C_2 \sin z]$

$= x[C_1\cos(\log x) + C_2\sin(\log x)]$.

$$P.I. = \frac{1}{D_1^2 - 2D_1 + 2} ze^z = e^z \frac{1}{(D_1+1)^2 - 2(D_1+1)+2} z$$

$$= e^z \frac{1}{D_1^2 + 1 + 2D_1 - 2D_1 + 2 + 2} z$$

$$= e^z \frac{1}{(1+D_1^2)} z = e^z (1+D_1^2)^{-1} z$$

$$= e^z [1 - D_1^2 + \cdots] z$$

$$= e^z \cdot [z - 0 + 0 - \cdots)$$

$$= e^z \cdot z = x \log x$$

So, General solution is

$$\boxed{y = x[C_1 \cos(\log x) + C_2 \sin(\log x)] + x \log x}$$

Ans //

⊛ **Legendre's Homogeneous linear differential Equation with variable coefficient OR OR equations reducible to homogeneous form with variable coefficient or Cauchy-Euler equation**

A D.E. of the form

$$(a+bx)^n \frac{d^n y}{dx^n} + a_1 (a+bx)^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_{n-1} (a+bx) \frac{dy}{dx} + a_n y = X$$

$a, b, a_1, a_2, \ldots a_n$ are constants.

X → Constant or function of x alone.

Can be reduced to homogeneous form by putting $a+bx = v$

$$\Rightarrow \boxed{b = \frac{dv}{dx}}$$

and $$\frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{dx} = b \frac{dy}{dv}$$

$$\boxed{\frac{d}{dx} \equiv b \frac{d}{dv}}$$

Similarly, $$\boxed{\frac{d^2y}{dx^2} \equiv b^2 \frac{d^2y}{dv^2}}$$

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Exp. Solve $[(3x+2)^2 D^2 + 3(3x+2)D - 36]y = 3x^2 + 4x + 1$ ——— (1)

Solution :→ Let $3x+2 = v$

so, $\frac{dy}{dx} = \frac{dy}{dv}\frac{dv}{dx} = 3\frac{dy}{dv}$

and $\boxed{\frac{d^2y}{dx^2} = 9\frac{d^2y}{dv^2}}$

so equation (1) becomes,

$v^2 \cdot 9\frac{d^2y}{dv^2} + 3v \cdot 3\frac{dy}{dv} - 36y = 3\left(\frac{v-2}{3}\right)^2 + 4\left(\frac{v-2}{3}\right) + 1$

$v^2\frac{d^2y}{dv^2} + v\frac{dy}{dv} - 4y = \frac{1}{27}(v^2-1)$ ——— (2)

which is homogeneous in y and v.

Let $v = e^z$ i.e. $\boxed{z = \log v}$, $D_1 \equiv \frac{d}{dz}$

so, (2) becomes

$D_1(D_1-1)y + D_1y - 4y = \frac{1}{2}\frac{1}{27}(v^2-1)$

$(D_1(D_1-1) + D_1 - 4)y = \frac{e^{2z}-1}{27}$

$(D_1^2 - 4)y = \frac{e^{2z}-1}{27}$, which is linear differential equation of second order with constant coefficient in y and z.

So, first we will find solution of this eqn.

A.E. $\Rightarrow m^2 - 4 = 0 \Rightarrow m = \pm 2$

so, C.F. $= c_1e^{2z} + c_2e^{-2z}$

$= c_1v^2 + c_2v^{-2}$

$= c_1(3x+2)^{+2} + c_2(3x+2)^{-2}$

P.I. $= \frac{1}{27}\frac{1}{D_1^2-4}(e^{2z}-1)$

$= \frac{1}{27}\left[\frac{1}{(D_1-2)(D_1+2)}e^{2z} - \frac{1}{D_1^2-4}e^{0z}\right]$

$= \frac{1}{27}\left[\frac{1}{(D_1-2)}\frac{1}{4}e^{2z} - \frac{1}{0-4}e^{0z}\right]$

$= \frac{1}{108}\left[ze^{2z} + 1\right] = \frac{1}{108}(ze^{2z}+1) = \frac{1}{108}(v^2\log v \cdot$

$= \frac{1}{108}\left[(3x+2)^2\log(3x+2) + 1\right]$

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Soln. (1) $(1+x)^2 \frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = 4\cos\log(1+x)$

Ans. $y = C_1 \cos\log(1+x) + C_2 \sin\log(1+x) + 2\log(1+x)\sin\log(1+x)$

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